



The impact of anti-organized crime on business environment: Evidence from China's anti-Mafia campaign

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ABSTRACT

This paper investigates the impact of China's 2018–2020 anti-Mafia campaign on the business environment. Using a novel dataset on organized crime prosecutions across China's prefecture-level cities and above, we find that the anti-Mafia campaign significantly improved the overall business environment. Probing the mechanisms, we show that the anti-Mafia policy enhanced the business environment through four key channels: increasing international trade and investment, bolstering domestic commercial activities, improving government effectiveness, and promoting public safety. Our findings pass a battery of robustness checks. We contribute to studies on organized crime economics and highlight the need for continued efforts to combat the Mafia.

1. Introduction

The economic impact of organized crime has attracted significant attention from crime economists over the past decade due to the escalating global risks and economic losses associated with such activities (Global Organized Crime Index 2023, n.d.). However, existing research in this field has disproportionately focused on the Italian Mafia (Calamunci & Drago, 2020; Daniele & Marani, 2011; Detotto & Pulina, 2013; Mirenda, Mocetti, & Rizzica, 2022; Mocetti & Rizzica, 2023; Neanidis, Rana, & Blackburn, 2017; Pinotti, 2015b) and organized crime in certain Latin American and West African countries (Mazzitelli, 2007; Mohammed, Hossein, & Nwokolo, 2022; Soler, 2012). Empirical studies on China, a country that have been proactive in combating organized crime in recent years,¹ are notably scarce. This lack of research raises doubts about the applicability of widely validated theories, particularly those suggesting that anti-Mafia efforts directly benefit the economy, to the Chinese context (Wang, 2019). Currently, it is not yet clear how Mafia

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¹ “Mafia” refers to organized crime groups in Italy and shares key elements with the targets of China's anti-gangsterism campaign, including: (a) involving three or more people; (b) requiring a certain level of organization and the commitment of more than one crime; (c) employing violence, intimidation, or other means of direct control or indirect suppression; and (d) generally aiming to obtain economic benefits or exert other illegal influence. The difference is that China's anti-gangsterism campaign not only fights against traditional Mafia-like organizations but also groups labeled as “evil forces.” These groups are defined as collections of individuals, typically three to ten, who repeatedly engage in illegal activities within a specific industry or region but have not yet formed a mature and stable criminal organization. In this paper, the terms “anti-organized crime,” “anti-Mafia,” and “anti-gangsterism” are used interchangeably to describe efforts to combat organized crime. See China's *Anti-organized Crime Law* (2021) and Article 416-bis of the Criminal Code of Italy (Criminal Code, 1996).

activities or anti-Mafia campaign will impact the Chinese economy.

From 2018 to 2020, China undertook one of the world's largest anti-gangsterism operations, the "Sweep Away Black, Eliminate Evil (*Saohei Chu'e*)" campaign. This campaign was initiated by China's most central authorities, mobilizing a large-scale effort involving the police, procuratorates, courts, party organizations, government, and the public to detect and arrest Mafia members. Moreover, agencies at all levels nationwide are required to specifically report on the implementation of this campaign in their annual work reports. Ultimately, it results in the arrest of more than 240,000 suspects allegedly involved in gang-related crimes (Yang, 2021), the vast majority of whom were eventually convicted.

Overall, China's specialized anti-Mafia campaign has been a tremendous success. The data in Fig. 1, showing a more than 70% decrease in prosecutions of Mafia members post-campaign, indicates that the movement has apprehended the majority of organized criminals in China. In addition, the substantial socio-economic and political differences among Chinese cities enable an examination of the varied economic impacts of China's anti-Mafia campaign across diverse urban settings. To move beyond many previous studies in the Mafia economics research that focused on macro-level indicators such as economic growth, we investigate micro-level market performance, specifically the business environment.

Treating the anti-Mafia policy as a plausibly exogenous shock to the business environment, we employ a difference-in-differences (DID) approach to examine the impact of the anti-Mafia campaign on the business environment of all Chinese cities at the prefecture level and above. Specifically, we take the year 2018 and onwards as the implementation years of the policy. We then assess the impact level of the anti-Mafia policy on each city by calculating the ratio of the average number of individuals prosecuted for Mafia-type crimes during the three-year period of this campaign (2018–2020) to the city's average annual permanent population within the same timeframe. The city's Business Environment Index² is derived from the "China City Business Environment Database 2023," which evaluates the business environment in cities across four dimensions: market, governmental, legal, and cultural environments, covering the period from 2017 to 2021. We choose this index as a measure of the business environment for two main reasons. Firstly, it is recognized for its authority and credibility, making it widely cited in research on China's business environment. Secondly, this database provides data at the prefecture-level, which offers a more detailed perspective compared to other indicators only at the provincial level.

Our findings indicate that the anti-Mafia campaign from 2018 to 2020 significantly improved the business environment. Our analysis has passed the parallel trends test, and these findings remain robust following a battery of robustness checks, including a placebo test, propensity score matching–difference in differences (PSM–DID), an alternative measure for the explanatory variable, additional fixed effects, sensitivity analysis, and excluding the impact of extreme outliers. The mechanism analysis reveals that the anti-Mafia campaign enhanced international trade and investment, bolstered domestic economic activities, improved the governmental environment, and ameliorated public safety. These findings suggest that the anti-Mafia campaign generates a comprehensive governance effect, which collectively contributes to optimizing the overall business climate in China.

We make contributions to two bodies of literature. First, we provide new evidence and mechanism interpretation from China on the study of the economic impact of anti-Mafia activities, enriching the literature in this field. Previous research has primarily demonstrated the negative economic impacts of Mafia organizations in Italy and South American countries, examining various individual indicators such as regional economic growth, per capita income, business operations, corporate innovation, market competition, private investment, human capital, public employment, and the distribution of public resources (e.g., Asmundo & Lisciandra, 2008; Benton, 2017; Caglayan, Flamini, & Jahanshahi, 2021; Calamunci & Drago, 2020; Daniele & Marani, 2011; Detotto & Pulina, 2013; Ferrante, Fontana, & Reito, 2021; Mocetti & Rizzica, 2023; Operti, 2018; Pinotti, 2015a, b; Ramos & Ashby, 2013; Soler, 2012). Our paper complements this literature by providing robust evidence of how China's anti-Mafia policy serves as an exogenous shock that improves the overall business environment in cities. Second, at the data level, our research goes further than previous studies that used provincial-level data (Le Moglie & Sorrenti, 2022; C. Liu, Wang, & Peng, 2023; Z. Li & Cheng, 2024) by employing more precise city-level data. The prosecution data used in our study is also more scientifically grounded than the conviction data used in prior research (Pinotti, 2015a; Neanidis et al., 2017; Z. Li & Cheng, 2024). Finally, in terms of policy significance, the 2018–2020 anti-Mafia campaign represents China's fifth and likely final large-scale crackdown on organized crime, considering the subsequent enactment of the Anti-Organized Crime Law. Our research and policy recommendations offer timely empirical justification and suggestions for improving this legislation.

The reminder of the paper is structured as follows. In Section 2, we introduce the institutional background and develop two competing, testable hypotheses based on alternative theoretical perspectives. Section 3 presents our data sample and variables. Section 4 presents the empirical analysis. We introduce the empirical model, discuss the main regression results, and conduct a series of robustness checks to ensure the credibility and validity of our findings. Section 5 explores the mechanisms driving the impact of anti-Mafia campaign on the business environment by analyzing changes in the sub-indicators of the business environment index following the implementation of the anti-Mafia policy. Section 6 concludes the paper and offers policy recommendations to optimize the future anti-organized crime efforts in China.

² See Appendix A for details.

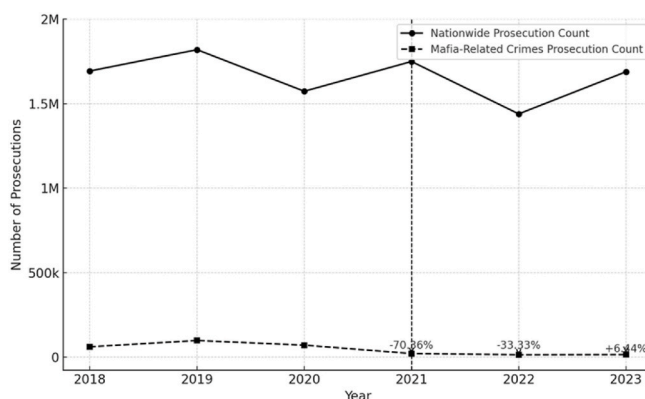


Fig. 1. Nationwide prosecution trends (individuals, 2018–2023).

2. Institutional background and hypothesis development

2.1. Institutional background

In 2018, China initiated a nationwide anti-Mafia policy, launching the three-year “Sweeping Black and Eliminating Evil” campaign. This initiative represented a concerted effort by the Chinese government to crack down on Mafia-related associations, aiming to secure the social and economic environment across the country. The 2018–2020 anti-Mafia campaign represents not only one of the largest anti-Mafia initiatives in Chinese history but also ranks among the most extensive globally, at least in terms of the number of people arrested. Moreover, with China implementing a normalized anti-gangsterism policy and enacting the ([Anti-organized Crime Law, 2021](#)) starting from 2021, the 2018 campaign may very well be the last concentrated, campaign-style law enforcement effort against Mafia groups in China.

The anti-Mafia policy received high-level political and judicial endorsement from the very beginning, as well as substantial financial and personnel backing. On January 24, 2018, the Central Committee of the Communist Party of China (“CCP”) and the State Council issued the “Notice on the Special Campaign to Sweep Away Black Societies and Eradicating Evil Forces,” mandating a three-year special campaign against all Mafia groups in China (*Notice on the Special Campaign to Sweep Away Black Societies and Eradicating Evil Forces, 2018*). To implement the top authority’s decision, on February 5, 2018, the Supreme People’s Court, the Supreme People’s Procuratorate, the Ministry of Public Security, and the Ministry of Justice issued the “Notice on Severely Cracking Down on Crimes and Illegal Activities of the Black Societies and Evil Forces in Accordance with the Law” (*Notice on Severely Cracking Down on Crimes and Illegal Activities of the Black Societies and Evil Forces in Accordance with the Law, 2018*), followed by the initiation of the campaign by their subordinate institutions. To prevent lax enforcement by local authorities, the central government dispatched several senior officials at the ministerial level to each province to supervise the implementation of the policy on three occasions: in July 2018, April 2019, and June 2019. The central CCP required local “political and legal committees”³ to provide financial, technical, and personnel support for the anti-gangsterism campaign to ensure that local authorities had sufficient resources to implement this policy (*National Plan for Supervision Work on the Special Campaign to Crack Down on Gangs and Eliminate Evil, 2018*). Several nationwide meetings and unified arrangements, such as “promotion meetings,” “retrospective inspections,” and “six-clearing actions,” were also held to continuously assess the progress of the anti-Mafia efforts and pressure local authorities to maintain their commitment.

Local authorities utilized four effective strategies to thoroughly implement the anti-gangsterism policy. First, they encouraged surrender. The joint notice issued by the four departments stated that Mafia members who voluntarily surrendered between February 5 and March 1, 2018, could receive lenient or mitigated punishments, whereas those who would not face more severe sanctions. Second, they carried out massive anti-gangsterism propaganda and mobilization. For instance, conspicuous red banners were hung in various urban and rural settings, such as schools, workplaces, train stations, communities, and entertainment venues, urging citizens to participate in the anti-gangsterism campaign ([Zang & Zou, 2022](#)). This approach not only conveyed political pressure to local officials but also helped to build a societal consensus against Mafias. Third, the anti-Mafia work was set as a performance evaluation indicator for law enforcement and judicial personnel to motivate them to actively implement anti-gang measures ([R. Xu & Liu, 2022](#)). Fourth, they encouraged peer competition. Starting from 2018, national police offices and judicial agencies added a dedicated section on the “Sweep Away Black, Eliminate Evil” campaign in their annual work reports, and combating more Mafia members would be publicized as an achievement.⁴

With the combined efforts of the central and local governments, China’s anti-Mafia policy led to substantial outcomes. From 2018

³ The Political and Legal Committee, or known as “Zhengfawei,” is a crucial institution of the CCP that oversees the political and legal systems to ensure alignment with party policies. It coordinates the work of legal enforcement and judicial agencies to enforce CCP directives.

⁴ For example, the Wuxi Municipal Procuratorate wrote in its 2018 work report: “the number of prosecuted [Saohei Chu’e] cases ranks among the top in the province.”

to 2020, China cracked down on 3644 black societies, 11,675 “evil forces,” prosecuted around 230,000 Mafia suspects, and over 43,000 Mafia members voluntarily surrendered (J. Zhang, 2021). In the wake of the policy, the number of Mafia members prosecuted by national procuratorates dropped sharply, falling to less than 30% of the number prosecuted in 2020. Therefore, the 2018–2020 anti-Mafia policy can be regarded as effective in rooting out the majority of Mafia members in China,⁵ and the special reports on this campaign by the national prosecution offices have become one of the best datasets for studying Chinese Mafia.

2.2. Hypothesis development

The impact of anti-organized crime policies on the business environment is complex and multifaceted. It may not manifest as a straightforward positive effect as might be intuitively perceived by the general public, especially over shorter time periods. Therefore, we establish two competitive hypotheses. On the one hand, the anti-Mafia policy and the ensuing crackdowns on Mafia-related crimes can contribute to better social governance and stability, thereby improving the overall business environment in a region (Ferrante, Reito, et al., 2021). On the other hand, the short-term, campaign-style enforcement can represent a sudden disruption to existing business models and underground power structures (Ghosal & Ye, 2019; Chen, Fan, Wang, & Wang, 2024). These abrupt changes may destabilize the business landscape and underground order, creating an unstable environment that is detrimental to business operations over a short period.

The anti-Mafia policy can positively influence the business environment for four primary reasons. First, the anti-Mafia campaign could increase a city's appeal to international trade and investment. Foreign trade and investment not only provide essential capital and technology but also contribute to improved productivity and overall economic development, particularly in developing countries that heavily rely on foreign investors (Amirahmadi & Wu, 1994; Hsiao & Shen, 2003; Tarzi, 2005). However, the high incidence of organized crime has been shown to diminish a region's attractiveness for international investment by signaling higher business costs, social instability, and political uncertainty (Ashby & Ramos, 2013; Daniele & Marani, 2011; Garriga & Phillips, 2023). The introduction of anti-Mafia policy in these regions has the potential to mitigate these concerns that deter international business professionals and encourage increased investment, thereby improving the local business environment.

Second, the anti-Mafia policy has the potential to stimulate domestic commercial activities, including finance, entrepreneurship, innovation, and business transactions. Existing research indicates that traditional Mafia groups often plunder legitimate assets from business owners through violence or threats (Lisciandra, 2014). For instance, on average, 58% of businesses in Sicily have been subject to extortion in the form of protection money, with these costs estimated to exceed 1.4% of the region's GDP (Asmundo & Lisciandra, 2008). In modern times, Mafias have been found to have extensively infiltrated in the legitimate economy through corporate structures (Mirenda et al., 2022). By colluding with local government officials and employing various illegal methods, they suppress private investment (Calamunci & Drago, 2020), weaken competition and entrepreneurship (Ferrante, Fontana, & Reito, 2021; Le Moglie & Sorrenti, 2022), hinder innovation (Caglayan, Flamini, & Jahanshahi, 2019), and create barriers to market entry (Slutzky & Zeume, 2023). Combating Mafia groups can eliminate the illegal ongoing costs and unfair competitive advantages imposed on businessmen, thereby freeing up more profit space for legitimate businesses and encouraging market financing, business startups, corporate innovation, and transaction activities, overall enhancing the business environment.

Third, the anti-Mafia policy may improve governmental environment. A very important feature of China's anti-Mafia movement is the simultaneous crackdown on the “protective umbrellas” of Mafia groups, namely corrupt officials who provide them with protection. During the anti-Mafia campaign, China's anti-corruption agencies dealt with a total of 68,100 such “protective umbrellas,” nearly 10,000 of whom were subjected to criminal prosecution (*The Three-Year Campaign*, 2021). The government, after the anti-Mafia policy, will be a cleaner and, theoretically, more efficient one (Montes & Paschoal, 2016; H. Zhang, An, & Zhong, 2019; Hao, Liu, Zhang, & Zhao, 2020). The governance effect of the anti-Mafia policy at the governmental level, in the long run, should be more beneficial for the business environment than those at the corporate level.

Fourth, the anti-Mafia policy exerts a significant deterrent effect not only on Mafia-related and corrupt activities but also on other types of crimes. For example, one year after the implementation of the anti-Mafia policy, national criminal cases decreased by 4.1%, severe violent crimes decreased by 10.3%, and gun-related cases decreased by 34.6% (*Striking Strongly at the Crucial Moment*, 2020). During periods of intensified enforcement, other criminals may reduce their criminal activities due to the prevailing harsh stance, which indirectly benefits public safety (La Spina, 2004), as well as the business environment.

Based on the above analysis, we propose our first hypothesis as follows:

H1A. The impacts of anti-Mafia policy will positively affect the business environment

In contrast, we speculate that the sudden anti-Mafia policy will disrupt the existing market order and underground power structures, thereby causing a short-term instability and negatively impacting the business environment. We propose this opposite hypothesis for the following three reasons.

Firstly, China's anti-Mafia campaign has sealed off, detained, and confiscated a vast array of Mafia-related assets, amounting to

⁵ We acknowledge that some scholars have suggested the COVID pandemic may have influenced criminal prosecutions in China. However, as seen in Fig. 1, although there are fluctuations in the nationwide prosecution data before and after the pandemic, the number of Mafia-related prosecutions has remained consistently low following the anti-Mafia campaign. This trend to some extent demonstrates the effectiveness of the campaign in apprehending the majority of Mafia members. The data were extracted from the annual work reports of the Supreme People's Procuratorate for the corresponding years, which can be accessed at <https://www.spp.gov.cn/spp/gzbg/index.shtml>.

hundreds of billions of RMB, dealing a heavy blow to Mafia organizations (The Three-year, 2021, The Three-Year Report Card for the Special Campaign, 2021). However, due to the extensive infiltration of China's Mafia groups into the legitimate economic sectors, illegal assets are often intermingled with legal ones, making it challenging for law enforcement officers to distinguish between them (J. Cai, 2021). Under the pressure of campaign-style enforcement, some officers fail to differentiate or incorrectly differentiate between mixed assets, even going as far as confiscating the properties of Mafia members' family members (Zeng & Chen, 2022). The impact of anti-Mafia policy on a portion of legitimate assets may, in the short term, affect the local business environment (Wang, 2019).

Secondly, aligned with Li and Cheng's (2024) hypothesis, the sudden implementation of the anti-Mafia policy initially lacked specificity regarding targeted industries or sectors and hence introduced a certain level of uncertainty to the market. It was only in the latter stage that the ten key sectors including construction, transportation, and natural resources was identified for focused investigation, based on the distilled experience from various locales. During this period, the dissemination of any information from the anti-Mafia campaign could potentially exacerbate the uncertainty regarding investor sensitivity in this economic climate. Through event study, Li and Cheng (2024) illustrate that the uncertainty effect engendered by the anti-Mafia policy leads to a short-term decline in asset prices, which might adversely affect the business environment.

Thirdly, the apprehension of "protective umbrellas" might disrupt the pre-existing underground power hierarchy, potentially triggering more violent incidents that could adversely affect the business environment in the short term. According to Chen et al.'s (2024) Godfather Theory and empirical results, some corrupt officials may act as arbitrators for local Mafia groups, much like a godfather. The loss of such arbitrators leads to an increase in violent confrontations among gang groups in the short term, which may disrupt local public security and thereby affect the business environment. A similar mechanism has been observed in Italy, where combating Mafia-related crime has led to an institutional vacuum, resulting in increased market uncertainty and reduced entrepreneurship rates in the affected regions (Operti, 2018). Overall, the crackdown on Mafia forces may shock the established interest landscape and market order, leading to market turmoil, heightened uncertainty, and a challenging business environment, especially in the short run. Hence, we put forward the following hypothesis:

H1B. The impacts of anti-Mafia policy will negatively affect the business environment

3. Data and variables

3.1. Data sources

Our research data are obtained from the following databases: work reports of procuratorates in Chinese prefecture-level cities and above, statistical yearbooks of each province, "China City Statistical Yearbook," and "China City Business Environment Database 2023." These datasets contain information on the number of people prosecuted for suspected Mafia-related crimes, permanent resident population, business environment index system, and other economic indicators for each city.

The work reports of procuratorates in Chinese prefecture-level cities and above provide information on the number of individuals prosecuted for suspected Mafia-related crimes in each city. We manually collect and compile the reported prosecution number from the work reports of procuratorates in 290 prefecture-level cities and above in China from 2018 to 2020.⁶ We cross-check this data with the case number involving Mafia-related crimes accepted by courts during the same period and removed 68 missing or conflicting data points, ultimately forming a dataset of Mafia infiltration containing 802 data points. The missing rates for the three years are 5.86%, 5.52%, and 12.07%, respectively, indicating a generally reasonable data structure.

Our business environment data is derived from the "China City Business Environment Database 2023" (S. Zhang, Zhang, & Huang, 2023). This database contains 4 first-level indicators, 15 s-level indicators, and 24 third-level indicators. It provides a scientific quantitative assessment of the business environment of all Chinese cities at or above prefecture-level from 2017 to 2021, based on four major aspects: market environment, governmental environment, legal environment, and cultural environment. It is widely accepted as an authoritative source of business environment metrics in China.

The statistical yearbooks of each province and the "China City Statistical Yearbook" provide multi-faceted statistical indicators at the prefecture-level city level, including economic, financial, population, education, and scientific research metrics. The control variables used in this study are primarily sourced from these statistical yearbooks. The specific sources of each control variable are detailed in the variable construction section.

3.2. Variables construction

Measuring the policy impact as the independent variable. We use the ratio of the average number of individuals prosecuted during the city's three-year anti-Mafia campaign to the city's average permanent population over the same period to represent the intensity of a

⁶ While Chinese official data might be suspected of being less complete and accurate than international data, only local data is available for this issue, and we believe our data is relatively reliable. First, the number of Mafia-related prosecutions is not sensitive data. It does not imply any value judgments but is simply an individual count of prosecutions. Second, higher-level statistics are more likely to be manipulated, but our data is at the city level, which is more robust than provincial or national-level statistics. Third, we did not use the data directly as found; instead, we conducted cross-validation and verified the accuracy of the data through the number of cases handled by the local courts. We did find some contradictory data, such as the number of gang-related cases handled by the courts being much higher or lower than the number of prosecutions claimed by the procuratorate. We have excluded such data from our analysis.

city's exposure to the 2018–2020 anti-Mafia policy (*API*). Firstly, opting for the number of prosecutions rather than the number of convictions is justified. According to the Criminal Funnel Theory, the criminal justice system is a funnel-shaped structure from left to right, where “crimes” flow through a series of structured filtering mechanisms, and a certain amount of “crimes” are filtered out at each stage of victim reporting, police investigation, prosecutor prosecution, and court sentencing (Mayeux, 2018). Therefore, data from later stages are more distorted, and using conviction data could underestimate the quantity of crime due to factors such as procedures, evidence, trial duration, and confidentiality requirements. Secondly, it is also rational not to use the number of victim reports and police case filings. This campaign's search for Mafia members has shifted from reactive to more proactive investigations (Yin & Mou, 2023), which may result in fewer cases being reported by the public but more cases being actively investigated by the police. Not using the number of police case filings is due to concerns over excessive law enforcement in the anti-gangsterism campaign (D. Xu & Shi, 2019). Relying on police investigation data might overestimate the number of Mafia members in a city. China's procuratorates, as legally neutral supervisory bodies (Lin & Wu, 2022), perform a basic function of case quality control. Although prosecutors are also encouraged to prosecute more Mafia members, correcting wrongful cases is among their performance evaluation indicators and report content. Considering criminological theory and the realities of anti-Mafia enforcement, we argue that examining the number of individual prosecutions for Mafia-related crimes is a rational choice for our study. To capture the impact of the issue of the anti-Mafia policy, we construct a dummy variable *post*, which is assigned a value of 1 in the implement year of the anti-Mafia policy, and 0 otherwise.

Measuring the business environment as the dependent variable. We employ the Business Environment Index constructed in the “China City Business Environment Database 2023” to measure the level of the business environment in each city. This index evaluates the business environment in cities from four aspects: market, governmental, legal, and cultural environments, and ultimately provides data on the city business environment index for 296 prefecture-level and above cities in China from 2017 to 2021.

Control variables. The core explanatory variables and the dependent variables in this study are measured at the city-year level. Although we have introduced city fixed effects and year fixed effects in the fixed effects model to control for the impact of indicators that vary among city entities and those that vary among years, there may still exist other factors varying at the city-year level that influence the business environment. Therefore, we further incorporate the following control variables into our model to mitigate estimation biases caused by omitted variables as much as possible: (1) the level of economic development, measured by the GDP growth rate (*GDPYoY*); (2) the degree of financial development, gauged by the ratio of the year-end balance of loans from financial institutions to GDP (*credit*); (3) the fiscal capacity, measured by the proportion of local government budgetary expenditures to GDP (*govexp*); (4) the development of education and science, assessed by the ratio of local government expenditures on education and science to total budgetary expenditures (*edufis*, *scifis*); (5) the market structure, evaluated by the number of industrial enterprises above a designated size (*logarithmically transformed*, *lnfirms*); (6) the quality of human resources, measured by the ratio of the number of students in regular higher education institutions to the average permanent population (*studentr*); (7) the standard of living, measured by the average wage of employees (*logarithmically transformed*, *lnwage*). The data of control variables are obtained from the “China Stock Market & Accounting Research (CSMAR)” database and the “China City Statistical Yearbook.”

4. Empirical results and analysis

4.1. Empirical model

To identify the impact of the anti-Mafia policy on the business environment, we refer to the research of Xiong, Huang, Yang, and Wang (2024), and adopt a Two-Way Fixed Effect DID regression model to examine our research hypotheses:

$$BE_{it} = \alpha + \beta API_i \times post_t + \Phi Controls + \gamma_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where *i* denotes the prefecture-level and above city, and *t* represents the year. The dependent variable *BE_{it}* represents the business environment, measured by the business environment index from the “China City Business Environment Database 2023.” The core explanatory variable *API_i* represents the anti-Mafia policy impact, measured by the ratio of the average number of people prosecuted for suspected Mafia-type crimes in a city from 2018 to 2020 to the city's average permanent population over the same period. Choosing to use the three-year average number of prosecutions instead of examining each year in detail is because organized crime and its investigation and prosecution are long-term processes that may span across years. Using a three-year average can better avoid errors due to prosecution delays. We use *post_t* to capture the impact of the implementation of the anti-Mafia policy, where *post_t* is a dummy variable assigned a value of 1 from 2018 to 2021, and 0 for 2017. *Controls* is a set of control variables that include various characteristics of prefecture-level and above cities measured at the city-year level. These variables aim to account for additional factors that may affect the city's business environment and are not addressed by fixed effects. γ_i represents city fixed effects, δ_t represents year fixed effects, and ε_{it} denotes the error term.

We are particularly interested in the coefficient β . A positive β means that cities more heavily affected by anti-Mafia policy tend to have a better business environment, indicating that we can accept the first hypothesis (H1A). Conversely, a negative β suggests that cities more deeply impacted by anti-Mafia policy have a worse business environment, allowing us to accept the second hypothesis (H1B).

4.2. Baseline results

We present the results of Equation (1) in Table 1. Specifically, column (1) displays the estimation results, including the explanatory variable and control variables in the regression model. Columns (2) and (3) further control for city fixed effects and year fixed effects, respectively. Column (4) lists the estimation results, controlling for both city and year fixed effects. The coefficients of the interaction term $post \times API$ are consistently positive across columns (1)–(4) and significant at the 5% level, suggesting that the anti-Mafia policy introduced in 2018 helps improve the business environment.

To eliminate the impact of the COVID pandemic, columns (5)–(8) present results excluding samples from 2020 and beyond. Column (5) includes only the explanatory and control variables, while column (6) introduces the city fixed effects based on column (5). Column (7) introduces the year fixed effects, and column (8) controls for both the city and year fixed effects. After accounting for the impact of the COVID pandemic, the coefficients of the interaction term $post \times API$ become significantly positive at the 1% level. This indicates that, before the pandemic, the positive effect of the anti-Mafia policy on the business environment was even more pronounced. The findings support H1A.

4.3. Robustness checks

4.3.1. Parallel trend test

DID regression models rely on the parallel trend assumption. Following Beck, Levine, and Levkov (2010), D. Liu, Zhou, Chen, Wan, and He (2021), T. Li, Lu, and Wang (2023), and Xiong et al. (2024), we construct the model below to perform a parallel trends test on our samples:

$$BE_{i,t} = \alpha + \sum_j \beta_j CGC_i \times year_t^j + \Phi Controls + \gamma_i + \delta_t + \varepsilon_{i,t}, \quad (2a)$$

where $year_t^j$ is a dummy variable used to indicate different years, with j ranges from 2017 to 2020. For instance, $year_t^{2018}$ represents whether the sample belongs to the year 2018. It is assigned a value of 1 if the sample is from 2018, to 0 otherwise. Since that 2018 is the year of policy implementation, we would expect the coefficient of the interaction term for 2017 to show no significant difference from 0. The regression coefficients, displayed in Fig. 1, reveal that the coefficients before and during the year of policy implementation are

Table 1
Baseline regression results.

	Dependent Variable: City Business Environment Index (BE)							
	Full Sample				Excluding Samples from 2020 and beyond			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>post</i>	−1.3810*** (0.4270)	−0.2755 (0.3427)			−1.6459*** (0.3850)	−0.1432 (0.4293)		
<i>API</i>	−0.6873*** (0.1858)		−0.6894*** (0.1873)		−0.7771*** (0.2045)		−0.7743*** (0.2060)	
<i>post × API</i>	0.3539** (0.1738)	0.3694** (0.1807)	0.3675** (0.1813)	0.3669** (0.1820)	0.6128*** (0.1900)	0.5078*** (0.1703)	0.6212*** (0.1964)	0.5161*** (0.1767)
<i>GDPYoY</i>	−0.0883* (0.0466)	−0.1410*** (0.0452)	−0.1584** (0.0634)	−0.1264** (0.0589)	−0.1112 (0.0842)	0.0913 (0.0993)	−0.1365 (0.0847)	0.0704 (0.0989)
<i>credit</i>	0.5391* (0.2879)	0.1419 (0.2103)	0.5126* (0.2923)	0.1879 (0.2112)	0.4825* (0.2714)	−0.2944* (0.1597)	0.4554* (0.2597)	−0.1895 (0.1756)
<i>govexp</i>	−9.4464*** (2.4595)	−1.9046 (5.7699)	−9.1885*** (2.4465)	−1.7516 (5.7612)	−10.4264*** (2.5870)	−2.6508 (5.3416)	−10.2816*** (2.5647)	−2.7622 (5.3260)
<i>scifis</i>	62.0004*** (18.2410)	45.9651** (20.4362)	60.6889*** (18.1154)	48.6742** (20.4232)	57.9061*** (18.0474)	34.8792 (30.6982)	56.8197*** (17.9074)	37.7828 (31.0449)
<i>edufis</i>	−5.2604 (6.7936)	−7.7373 (11.0563)	−4.6741 (6.6699)	−9.7324 (11.1910)	−6.4055 (6.8498)	7.0918 (12.0761)	−6.7051 (6.8478)	4.8245 (12.1721)
<i>lnfirms</i>	1.4680*** (0.2946)	−1.6859 (1.1601)	1.4773*** (0.2897)	−1.7852 (1.2101)	1.1715*** (0.3102)	0.4757 (1.3612)	1.1748*** (0.3062)	0.7089 (1.3680)
<i>studentr</i>	0.0001 (0.0009)	−0.0011 (0.0041)	0.0001 (0.0009)	−0.0005 (0.0043)	0.0003 (0.0009)	0.0066 (0.0051)	0.0003 (0.0009)	0.0084 (0.0052)
<i>lnwage</i>	5.8979*** (1.7001)	−0.0030 (0.9998)	6.5512*** (1.9480)	0.4784 (1.0694)	7.4761*** (1.5893)	−0.7694 (2.4506)	8.0138*** (1.6234)	2.8972 (3.1729)
<i>Constant</i>	−35.8894* (18.7347)	49.4850*** (13.4956)	−44.0553** (21.5885)	44.5280*** (15.0738)	−50.6945*** (17.5565)	39.0885 (29.1808)	−57.5981*** (18.0540)	−3.4458 (37.7826)
City Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes	No	No	Yes	Yes
Adjusted R ²	0.50	0.73	0.51	0.73	0.52	0.76	0.53	0.76
Observations	963	926	963	926	689	674	689	674

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period.

not significantly different from 0. This finding indicates that there were no significant differences between the treatment and the control groups before the implementation of the policy. However, the coefficient becomes significantly different from 0 one year after the policy implementation, supporting the parallel trends assumption required for the DID method.

Following the methodologies outlined by P. Li, Lu, and Wang (2020), D. Liu et al. (2021), T. Li et al. (2023), and Xiong et al. (2024), we conduct placebo tests to examine potential bias in the estimated coefficients attributed to unobserved factors. Specifically, we randomly sample the interaction term between policy implementation and policy intensity 500 times, re-estimating the coefficient with each sample. The results, depicted in Fig. 2, indicate that the placebo test coefficients have a mean of 0 and follow a normal distribution. Furthermore, there are significant differences between these placebo test coefficients and the coefficient from the baseline regression (dashed line). These results from Fig. 2 lend support to the conclusion that the observed impact of the anti-Mafia policy on the business environment is not attributed to other unobservable factors.

4.3.2. Propensity score matching–difference in differences model

To address the issue of sample selection bias, we employ the PSM–DID model to reassess the impact of the anti-Mafia policy on the business environment. The baseline DID model in our study is a continuous DID, where we label the sample cities based on the extent to which they are influenced by the anti-Mafia policy (*API*). However, PSM requires the dependent variable to be a dummy variable. Therefore, following Xiong et al. (2024), we define cities with *API* larger than the median as the treatment group to conduct a nearest-neighbor propensity score matching (PSM), and use the PSM sample to re-estimate Equation (1). The results listed in Table 2 indicate that the baseline findings still hold after using the PSM method.

4.3.3. Alternative proxy of policy intensity

For the robustness test, we adopt an alternative measure of policy intensity: the ratio of the total number of prosecutions over a three-year period to the city's resident population (*API_sum*). We re-estimate Equation (1) using this new explanatory variable. The results of this DID regression, along with its parallel trends test, are presented in columns (1)–(2) of Table 3. These findings show that, even when employing this alternative measure of policy intensity, the coefficient of the interaction term between policy implementation and policy intensity remains significantly positive, aligning with our baseline regression results. Additionally, the sample successfully passes the parallel trends test. This consistency supports the robustness of our main findings.

4.3.4. Adding fixed effects

To mitigate potential bias from omitted variables and to better control for other factors that may influence the business environment in cities, we further include province-year fixed effects in the baseline regression and rerun the analysis. The results are presented in columns (3) and (4) of Table 3. Column (3) displays the regression results without control variables, while column (4) incorporates these variables. Both sets of results demonstrate that, even after controlling for province-year fixed effects, the coefficient of the DID interaction term remains significantly positive. This robustness analysis provides further evidence of the stability of our findings.

4.3.5. Sensitivity analysis

In order to minimize the bias caused by omitted variables, we control for city fixed effects and year fixed effects in the baseline regression, and further include province-year fixed effects in the baseline regression to test the robustness of the baseline results. In this section, we conduct a supplementary sensitivity analysis using techniques developed by Cinelli and Hazlett (2020) and Cellini (2021) to measure the sensitivity of the baseline results to potential omitted variable bias.

Fig. 3 presents the coefficients and t-values of the core explanatory variable when the unobserved confounder is 1–3 times stronger than the observed GDP growth (*GDPYoY*) covariate. The x-axis represents the partial R-squared of the omitted variables on the core explanatory variable with control variables and fixed effects included in the baseline regression, while the y-axis represents the partial R-squared of the omitted variables on the dependent variable when the core explanatory variable and control variables, as well as fixed effects, are included. The left graph shows the estimated results of the coefficients, while the right graph shows the estimated results of the t-values. Benchmarking against the observed confounders, even when the unobserved confounder is 3 times stronger than the observed GDP growth (*GDPYoY*) covariate, the coefficients of the core explanatory variable remain positive and statistically significant at the 5% level. This indicates that the influence of anti-Mafia policy on business environment is not overturned by the addition of missing variables.

4.3.6. Excluding the impact of extreme values in the sample

Column (1) of Table 4 displays the regression results with all variables winsorized at the 1% level. The coefficients of the core explanatory variable remain significantly positive, indicating that our main findings persist even after mitigating the influence of extreme observations.

We use data of cities at or above the prefecture-level in China in the baseline analysis, where there are significant differences between different prefecture-level cities. The unique characteristics of municipalities directly under the central government and provincial capitals may affect the overall performance of the sample, leading to bias in the regression results. Therefore, we exclude the municipalities directly under the central government and provincial capitals from the sample and re-estimate the baseline regression. The regression results are shown in columns (2)–(5), where the samples of the municipalities directly under the central government are excluded in columns (2)–(3), and the samples of provincial capitals and the municipalities directly under the central government are excluded in columns (4)–(5). Additionally, samples during the pandemic are further excluded in column (3) and column (5). The

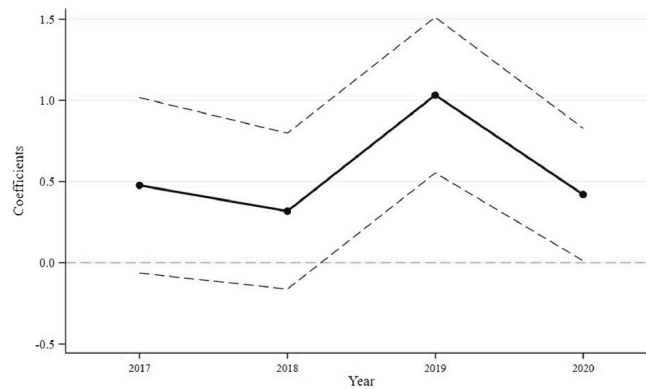


Fig. 2. Coefficient estimates and 95% confidence intervals of β_j 4.3.2. Placebo tests.

Table 2
Propensity score matching–difference in differences model.

Dependent Variable: City Business Environment Index (BE)	
(1)	
$post \times API$	0.3497* (0.1987)
Control Variables	Yes
City Fixed Effects	Yes
Year Fixed Effects	Yes
Adjusted R ²	0.72
Observations	921

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

Table 3
Alternative measure of anti-Mafia policy and addition of province-year fixed effects.

	Dependent Variable: City Business Environment Index (BE)			
	Alternative measure of API		Add province-year FE	
	DID	Parallel trend analysis		
	(1)	(2)	(3)	(4)
$post \times API_{sum}$	0.1037* (0.0553)			
$yearpre \times API_{sum}$		0.0384 (0.0778)		
$Yearpost1 \times API_{sum}$		0.2788*** (0.0728)		
$Yearpost2 \times API_{sum}$		0.1105* (0.0593)		
$post \times API$			0.5993* (0.3191)	0.6196* (0.3338)
Control Variables	Yes	Yes	No	Yes
City Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Province-year Fixed Effects	No	No	Yes	Yes
Adjusted R ²	0.73	0.73	0.75	0.73
Observations	926	926	1152	903

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

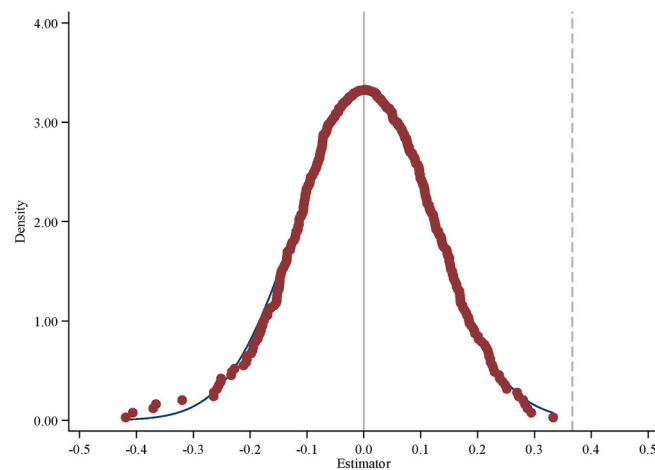


Fig. 3. Placebo tests results for the impact of anti-Mafia policy.

Table 4

Excluding the impact of extreme values in the sample.

	Dependent Variable: City Business Environment Index (BE)				
	winsorized at 1% level	excluding the municipalities directly under the central government		excluding the municipalities directly under the central government and provincial capitals	
		2017–2021	2017–2019	2017–2021	2017–2019
	(1)	(2)	(3)	(4)	(5)
<i>post</i> × <i>API</i>	0.5187** (0.2133)	0.3714** (0.1833)	0.5115*** (0.1756)	0.3179* (0.1770)	0.4269*** (0.1611)
Control Variables	Yes	Yes	Yes	Yes	Yes
City Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.72	0.70	0.72	0.65	0.67
Observations	926	911	663	820	596

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity. All variables winsorized at 1% level in column 1.

coefficients of the interaction terms in columns (2)–(5) are significantly positive, indicating that the main conclusions remain robust.

5. Mechanism analysis

The evidence presented thus far suggests that the anti-Mafia policy positively impacts the business environment. To further investigate the channels through which the anti-Mafia policy affects the business environment, we focus on four potential mechanisms based on the theoretical analysis outlined earlier: international trade and investment, domestic commercial activity, governmental environment, and social security. Fig. 4 illustrates these four mechanisms. We first examine the impact of anti-Mafia policy on four primary indicators of the Business Environment Index to identify the specific aspects through which the anti-Mafia policy affects the business environment. In addition, we also examine the impact of anti-Mafia policies on commercial activities and social security. The specific analyses of each mechanism test are as follows (see Fig. 5).

5.1. The impact of anti-Mafia policy on the four primary indicators of the business environment index

To investigate how China's anti-Mafia policy improves the business environment through various channels, we construct a specific model using the four primary indicators of the Business Environment Index as the dependent variables for regression analysis. The regression model is constructed as follows:

$$SubIndex_{i,t} = \alpha + \sum_j \beta_j CGC_i \times year_t^j + \Phi Controls + \gamma_i + \delta_t + \varepsilon_{i,t} \quad (2b)$$

The regression model is set up similarly to the baseline regression model, but the dependent variable is replaced with the four sub-indicators of the Business Environment Index. According to the description in the “China City Business Environment Database 2023,”

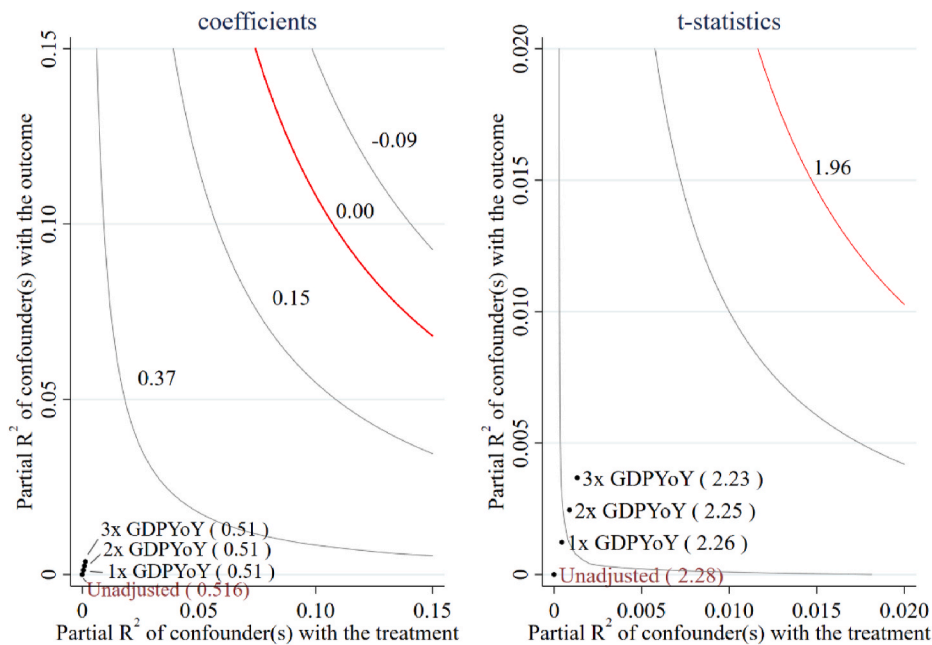


Fig. 4. Sensitivity analysis.

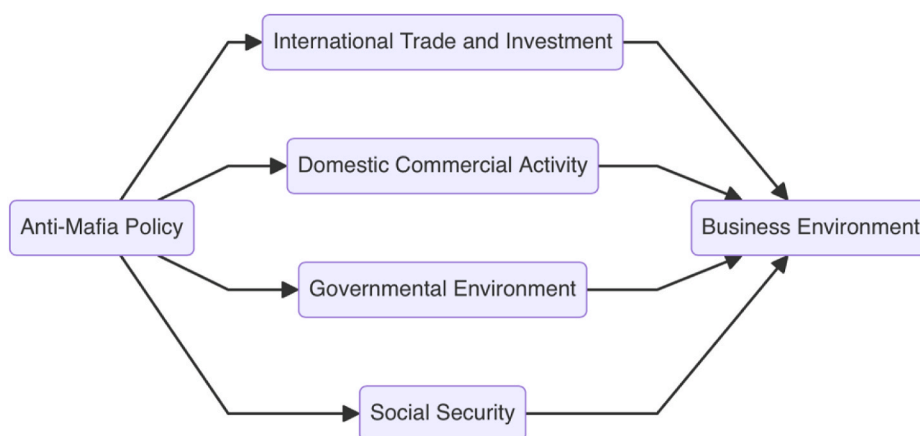


Fig. 5. Proposed mechanisms of anti-Mafia policy in improving the business environment.

the Business Environment Index consists of four sub-indicators based on the four aspects of business environment construction mentioned in the “13th Five-Year Plan⁷” of the country. These aspects are market environment, governmental environment, legal environment, and cultural environment. Since the cultural environment index primarily examines openness and inclusiveness, we primarily analyze this mechanism from the perspective of openness. The regression results using the four sub-indicators as dependent variables are presented in Table 5.

The results show that the anti-Mafia policy has a significant positive impact on the governmental environment and openness at the 10% and 5% significance levels, respectively. Regarding the governmental environment, the result indicates that China’s anti-Mafia policy can significantly improve government transparency and efficiency. It might be attributed to the fact that China’s anti-Mafia policy targets not only Mafia organizations but also their “protective umbrellas,” namely corrupt officials. Governments purged of corrupt elements are perceived as more transparent and efficient (Montes & Paschoal, 2016; H. Zhang et al., 2019; Hao et al., 2020),

⁷ The 13th Five-Year Plan (2016–2020) is a crucial economic development blueprint devised by the Chinese government. As an authoritative government document, it guides national resource allocation and reflects China’s long-term priorities, making it a pivotal reference for understanding China’s economy policies. This plan is often cited in academic and professional contexts to underscore the strategic direction of Chinese economic planning. See The 13th Five-Year Plan (2016).

Table 5

The impact of anti-Mafia policy on the primary indicators of the Business Environment Index.

	(1) <i>market</i>	(2) <i>government</i>	(3) <i>laws</i>	(4) <i>openness</i>
<i>post</i> × <i>API</i>	−0.1456 (0.0891)	0.7063* (0.3596)	0.1453 (0.1291)	0.4472** (0.1854)
Control Variables	Yes	Yes	Yes	Yes
City Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Adjusted R ²	0.95	0.48	0.51	0.77
Observations	926	926	926	926

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

conductive to a better business environment. As for the openness index, a detailed discussion will be presented in the following section.

In addition, we observe that the anti-Mafia policy did not exert a significant influence on the market and legal environments. The insignificant impact of anti-Mafia policy on legal environment may be attributed to the fact that the modifications to the laws and shifts in social disputes often require a longer period to manifest, and our short-term examination may not be sufficient to capture these changes. The stability of the market environment index could likely be attributed to the nature of its constituent indicators. Factors such as land supply, human capital, and network services tend to be unresponsive to the short-term policy changes, as they are influenced by a complex set of economic, social, and infrastructural conditions. Consequently, the brief observation period in our study may not capture the full extent of the anti-Mafia policy's influence on these indicators. Unlike long-term factors such as the legal environment and market infrastructure, commercial activities are more sensitive to the immediate effects of the anti-Mafia policy due to their strong sensitivity of capital. To better assess this potential impact, we develop a new indicator specifically focused on commercial activities, which encompasses key aspects such as financing, entrepreneurship, branding, and contract disputes. A detailed discussion of this re-estimation is presented in section 5.3.

5.2. The impact of anti-Mafia policy on international trade and investment

The anti-Mafia policy is believed to enhance the attractiveness of a region to international trade and investment (Daniele & Marani, 2008, 2011; Soler, 2012). Although laypeople might intuitively consider that foreign investors will not overly concern themselves with local organized crime when making investment decisions, details from reports on China's crackdown on Mafia groups, particularly in cities traditionally active in foreign trade, reveal a significant crowding-out effect of organized crime on foreign trade and investment. For instance, in the Chaoshan area of China, a hub for foreign investment, Meilai Shun Underwear City, which received over 500 million yuan in investment from Hong Kong businesses, was completely plundered by local Mafia groups who mobilized hundreds of members using violence and threats (A Major Mafia Case in Chaoshan! More Details Revealed, 2021). These Mafia groups were not completely dismantled until the anti-Mafia campaign. Another case occurred in Tianjin, the nation's leading city in automobile imports and exports. Local Mafias disrupted the local automotive import and export trade through extortion, racketeering, and other illegal means until the market normalized following the anti-Mafia efforts (Binhai New Area 2019) Sweeps Away Black and Evil Forces, 2019). These examples imply the disruptive impact of Mafia-related crimes on foreign trade, and thus, a significant increase in foreign trade can be anticipated following the anti-Mafia policy.

As shown in Table 5, the coefficient of the DID interaction term is significantly positive at the 5% level in the regression with openness as the dependent variable, indicating that China's anti-Mafia policy can significantly expand openness.

To further investigate the details of this mechanism, we refine the openness indicator into two components: international trade and international investment. We measure foreign trade by the ratio of total imports and exports to GDP (*EXIM*), foreign investment by the

Table 6

The impact of anti-Mafia policy on international trade and investment.

	(1) <i>EXIM</i>	(2) <i>FDI_firms</i>	(3) <i>FDI_contracts</i>
<i>post</i> × <i>API</i>	0.6824** (0.2675)	1.2879** (0.6345)	311.3919* (178.9246)
Control Variables	Yes	Yes	Yes
City Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Adjusted R ²	0.92	1.00	0.85
Observations	926	390	479

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

number of foreign-invested enterprises (*FDI firms*), and the number of contracts signed directly by foreign investors (*FDI contracts*). The regression results in Table 6 indicate that the positive effect of anti-Mafia policy on foreign trade, foreign-invested enterprises, and newly signed foreign contracts are significant at the 5%, 5%, and 10% levels, respectively. This suggests that the anti-Mafia policy positively enhances both foreign trade and foreign investment, thereby potentially benefiting the business environment.

5.3. The impact of anti-Mafia policy on domestic commercial activity

According to existing research, the impact of the anti-Mafia policy on domestic commercial activities is not as straightforwardly positive as its effect on attracting foreign investment; instead, it is complex and contentious. On the one hand, the policy facilitates the elimination of exploitation and the dismantling of monopolies, theoretically benefiting local economic activities such as financing, startups, innovation, and transactions (Caglayan et al., 2019; Ferrante, Reito, et al., 2021; Slutzky & Zeume, 2023). On the other hand, the sudden implementation of such policy can create market uncertainty due to its enforcement impact, thereby suppressing potential entrepreneurship and business activities (Operti, 2018). This market uncertainty could also lead to declines in stock and asset prices (Kelly, Pástor, & Veronesi, 2016; Bretscher, Hsu, & Tamoni, 2023; Z. Li & Cheng, 2024), which may deter some entrepreneurs from further engaging in commercial activities. Additionally, anti-Mafia measures may inadvertently force some legitimate businesses to exit the market (J. Cai, 2021).

The current indicator system of Business Environment Index lacks a primary indicator for commercial activity. To empirically test this issue, we identify four tertiary indicators most relevant to commercial activity from the existing business environment indicator framework based on theoretical considerations, including the financing level, the number of start-ups, the establishment of corporate brands, and contract dispute litigations. The increasing quantities of financing, start-ups, and corporate brands directly reflect the level of commercial activities. The selection of contract dispute litigation as an indicator is justified on the understanding that increased commercial activities imply a higher volume of transactions. Consequently, even if the litigation rate per transaction remains constant, an increase in total transactions will lead to more litigations in contract disputes (Ginsburg & Hoetker, 2006). We subsequently construct a new primary indicator based on the four tertiary indicators, i.e., the Commercial Activity (*ComActivities*). Our regression results in Table 7 on Commercial Activity are significantly positive at the 5% level, indicating that anti-Mafia policy has substantially increased commercial activities, thereby enhancing the business environment (see Table 8).

5.4. The impact of anti-Mafia policy on social security

There is a divergence of opinions in the existing literature regarding the impact of anti-Mafia policy on public safety, with two competing theories existing in this realm. The first is the Deterrence Theory. Beyond the severe strike on Mafia-related and corrupt activities, anti-Mafia policy may also deter other types of criminal behavior (La Spina, 2004). During periods of intensified enforcement against organized crime, other criminals may cease their illegal activities due to the heightened pressure on gang-related crimes, thereby improving public safety. The second theory is the Godfather Theory, which posits that anti-Mafia policy, by removing protective figures or “godfathers,” leads to a power vacuum within the local gang groups. As a result, this absence of a central arbitrator can provoke increased violent conflicts among the remaining members as they struggle for dominance.

To empirically examine the impact of anti-Mafia policy on public safety, we replace the dependent variable in the baseline regression with a public safety index ($SS_{i,t}$) and estimate the new regression model. The public safety index ($SS_{i,t}$) is constructed as follows.

$$SS_{i,t} = \frac{\max CC_t - CC_{i,t}}{\max CC_t - \min CC_t}$$

where $\max CC_t$ represents the maximum number of criminal cases per ten thousand people among all cities in year t , $CC_{i,t}$ represents the number of criminal cases per ten thousand people in city i in year t , and $\min CC_t$ represents the minimum number of criminal cases per ten thousand people among all cities in year t . A larger value of the public safety index ($SS_{i,t}$) indicates a higher level of public security. Therefore, we anticipate a positive regression coefficient for the interaction term $post \times API$ on the level of public security.

The findings show that the coefficient of the public safety index is significantly positive at the 5% level, indicating that anti-Mafia policy significantly reduces local criminal cases and enhances public safety levels, thereby potentially improving the business environment.

Our data challenges Chen's godfather theory, which empirically demonstrated that the removal of a “godfather” results in a significant 15% increase in organized violent criminal cases locally. Apart from differences in statistical methodologies, another possible explanation is the variation in contextual factors over time. The emergence of a power vacuum typically occurs only when there is insufficient government governance capacity (Operti, 2018). Chen's empirical data primarily illustrates that the anti-corruption campaign of 2013 led to a sharp surge in organized violence. In contrast, the governance capabilities of China's local authorities have been substantially strengthened since then (Wong, Tang, & Liu, 2020; C. Cai, Liu, & Jiang, 2023), reducing the likelihood of a power vacuum. Additionally, unlike parliament members who can be entirely infiltrated by Mafias in Italy (Acconcia, Corsetti, & Simonelli, 2014, p. 2186; De Feo & De Luca, 2017), Chinese officials' appointments do not require any services from Mafia groups, such as helping with elections, so there is no widespread collusion between Mafias and high-level officials but only possible connections with low-level police officers in China (Wang, 2017). The isolation of Mafias from the core power structures of local governments also helps prevent the emergence of power vacuums. In summary, the anti-Mafia policy has improved local public safety and

Table 7
The impact of anti-Mafia policy on domestic commercial activity.

	(1)
	<i>ComActivities</i>
<i>post</i> × <i>API</i>	0.8347** (0.3328)
Control Variables	Yes
City Fixed Effects	Yes
Year Fixed Effects	Yes
Adjusted R ²	0.92
Observations	926

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

Table 8
The impact of anti-Mafia policy on social security.

	(1)
	<i>SS</i>
<i>post</i> × <i>API</i>	0.3015** (0.1238)
Control Variables	Yes
City Fixed Effects	Yes
Year Fixed Effects	Yes
Adjusted R ²	0.95
Observations	926

Note: Robust standard errors clustered at the city level are in parentheses. * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level. To mitigate reverse causality issues, explanatory variables and all control variables are lagged by one period. Coefficients on Controls are omitted for brevity.

can positively impact the business environment.

Overall, through a more granular analysis of refined indicators of the Business Environment Index, coupled with existing theories outlined earlier, this section demonstrates that China's anti-Mafia policy is likely to improve the business environment through four primary channels: enhancing international trade and investment, strengthening domestic commercial activities, improving the governmental environment, and promoting social security.

6. Conclusion

This paper investigates the impact of China's 2018–2020 anti-Mafia campaign on the business environment in Chinese cities. Using a DID approach and a novel dataset on organized crime prosecutions and business environment metrics across China's prefecture-level cities and above, we find robust evidence that the anti-Mafia policy significantly improved the overall business environment. The findings are consistent across a battery of robustness checks. We propose four mechanisms through which the anti-Mafia efforts can enhance the business climate: 1) increasing international trade and investment by mitigating the risks and costs imposed by organized crime; 2) bolstering domestic commercial activities such as financing, entrepreneurship, innovation, and transactions by dismantling criminal monopolies and reducing extortion; 3) improving governmental transparency and efficiency by purging the “protective umbrellas” of Mafia groups; and 4) promoting public safety by deterring criminal activities and violence.

Our research makes several contributions. To the best of our knowledge, this is the first empirical study to document the economic impacts of China's recent anti-Mafia efforts using granular city-level data. The findings enrich the literature on organized crime economics, which has predominantly focused on the Italian context, by providing new evidence and insights from contemporary China. Methodology, the introduction of criminological theories also advances the field's research by guiding the selection of more scientifically valid data, such as prosecutorial indictments rather than trial conviction or policy investigation data. To conclude, our research underscores the importance of combating Mafias to foster a healthy business environment.

Data availability

Data will be made available on request.

Funding information

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Author statement

Xifen Lin: Conceptualization, Formal analysis, Writing-review & editing, Supervision, Funding acquisition. Wanqiang Wu: Conceptualization, Formal analysis, Writing-original draft. Huanqi Lan: Conceptualization, Formal analysis, Data curation, Methodology, Writing-review & editing. Yong Ma: Conceptualization, Formal analysis, Methodology, Writing-review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Business Environment Index

Primary Indicators	Secondary Indicators	Tertiary Indicators
Market Environment	Financing Innovation Fair Competition Resource Acquisition	Financing Level Number of Patents R&D Investment Establishment of Enterprise Brands Number of Start-up Enterprises Water Supply Electricity Supply Land Supply Human Capital Network Transportation Services Rental and Commercial Services
Government Affairs Environment	Market Intermediaries Government Care Government Efficiency	Degree of Government Care Government Scale Government Size E-government Level Government Integrity Degree
Legal Environment	Government Integrity Property Rights Protection Public Security Judicial Services	Patent Dispute Administrative Ruling Criminal Cases per 10,000 People Law Firms
Cultural Environment	Opening Up Social Credit	Foreign Trade Dependence Proportion of Foreign-invested Enterprises Contract Disputes

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